

Lecture 10 - Shadows

This lecture explains shadows as a visibility problem from the light source. A point is lit if the light can see it; it is shadowed if another object blocks that visibility.

Source chunk	course_text_parts/03_lectures/10_shadows.txt
Extracted slide pages	60
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Slide walkthrough

Each row gives a short reading comment for one extracted slide page. Use it to turn sparse slide bullets into a human-readable path through the lecture.

Page	Slide title	How to read it
1	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
2	Shadow Perception 1/3	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
3	Shadow Perception 2/3	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
4	Shadow Perception 3/3	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
5	Shadow Atmosphere	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
6	■ Shadows create a certain atmosphere	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
7	Shadow Atmosphere	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
8	Overview	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.

Page	Slide title	How to read it
9	10.1 Definitions	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
10	Classification of scene objects	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
11	Umbra and Penumbra	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
12	Hard and Soft Shadows	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
13	Description of Shadows	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
14	■ Shadows can be described differently	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
15	Observations 1/2	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
16	Observations 2/2	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
17	Shadow Calculation Methods	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
18	10.2 Ground Plane Shadows	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
19	Static Shadow Polygons	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
20	Render Static Shadow Polygons 1/2	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
21	Render Static Shadow Polygons 2/2	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
22	Static Shadow Polygons	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
23	Dynamic Shadow Polygons	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.

Page	Slide title	How to read it
24	Shadow Projection	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
25	Rendering Dynamic Shadow Polygons	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
26	10.3 Light Maps	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
27	Light Maps	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
28	Soft Shadows with Light Maps 1/2	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
29	Soft Shadows with Light Maps 2/2	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
30	10.4 Shadow Volumes	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
31	Shadow Volumes	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
32	Initial Considerations	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
33	Shadow Volume Geometry	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
34	Shadow Volume Realization	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
35	Shadow Volume Properties 1/2	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
36	Shadow Volume Properties 2/2	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
37	Stencil Shadow Volume	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
38	Implementation	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.

Page	Slide title	How to read it
39	Special Case: COP in the shadows	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
40	Shadow Volume Example	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
41	Shadow Volume Conclusions	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
42	10.5 Shadow Maps	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
43	Shadow Mapping	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
44	Projected Shadows	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
45	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
46	Depth Comparison	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
47	Coordinate Systems	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
48	Pseudo Code	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
49	Properties	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
50	View Volume Problem	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
51	Bias Problem 1/2	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
52	Bias Problem 2/2	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.
53	Aliasing Problem 1/3	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.

Page	Slide title	How to read it
54	Aliasing Problem 2/3	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
55	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
56	Aliasing Problem 3/3	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
57	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
58	■ Shadows in Visual Perception	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
59	Literature and other sources used in this chapter	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
60	■ Text Books	Read this as light-space visibility. Ask what blocks the light, what representation stores that information, and which artifact can appear.

10.1 Definitions

Source location: Section 10.1

Commentary

A shadow is not simply a dark texture. It is evidence that light visibility is blocked. Important distinctions include hard versus soft shadows, umbra versus penumbra, and geometric versus precomputed approaches.

For real-time graphics, shadows are usually approximated. The approximation must answer a visibility question cheaply enough for interactive rendering.

Mental model

Shadows are light-space visibility made visible in the camera image.

Check yourself

Why is shadow computation related to visibility determination?

10.2 Ground Plane Shadows

Source location: Section 10.2

Commentary

Ground plane shadows project an object onto a receiving plane. This is simple and can look convincing in restricted scenes, but it assumes a known planar receiver and does not handle general shadowing between

arbitrary objects.

The method is useful pedagogically because it makes the geometric nature of shadows explicit: a light source, an occluder, and a receiver define where the shadow lands.

Mental model

A planar shadow is projected geometry on a known receiver.

Check yourself

What scene limitation makes ground plane shadows less general than shadow maps?

10.3 Light Maps

Source location: Section 10.3

Commentary

Light maps store precomputed lighting or shadow information in textures. They can be efficient at runtime, but they are limited when lights, objects, or geometry move dynamically.

This is another example of a recurring graphics tradeoff: precompute for speed, but lose flexibility. Real-time rendering often mixes precomputed and dynamic techniques.

Mental model

A light map spends storage and preprocessing to save runtime shading work.

Check yourself

Why are light maps less suitable for fully dynamic moving lights?

10.4 Shadow Volumes

Source location: Section 10.4

Commentary

Shadow volumes construct the volume of space blocked from a light by an occluder. The camera view can then determine whether visible points lie inside that volume. Stencil buffer techniques are often associated with this method.

The strength is geometric precision for hard shadows. The cost is handling silhouette edges, volume construction, and robust stencil operations. It is a good contrast to shadow maps, which are image-based from the light view.

Mental model

A shadow volume is the 3D region where the light cannot reach.

Check yourself

Why are silhouette edges important for constructing shadow volumes?

10.5 Shadow Maps

Commentary

Shadow mapping renders the scene from the light's point of view and stores depth. During the camera pass, a surface point is transformed into light space and compared against the stored depth. If it is farther than what the light saw, it is in shadow.

This method is practical and widely used, but it has artifacts. Shadow acne comes from precision/self-comparison issues. Bias can reduce acne but too much bias causes detached shadows. Resolution, filtering, and light projection strongly affect quality.

Mental model

Shadow maps reuse the depth-buffer idea, but from the light's camera.

Check yourself

What exactly is stored in a shadow map?

End-of-lecture summary

If you remember only one thing from Lecture 10, remember this: This lecture explains shadows as a visibility problem from the light source. A point is lit if the light can see it; it is shadowed if another object blocks that visibility.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.